

Improved Delayed Neutron Group Parameters for Molten Salt Reactors

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Delayed neutron precursors (DNPs) emit delayed neutrons

Delayed neutron precursors (DNPs) are nuclides that undergo β^- decay, producing a daughter, called an emitter, which emits a delayed neutron [21].

Relevant data for DNPs includes:

$P_{n,i}$ = the emission probability [#]

τ_i = the half-life of DNP i [s]

$\chi_{d,i}$ = the delayed neutron energy spectrum from DNP i [-]

N_i = the number of atoms of i [#].

Accurately modeling the DNPs is important for transient simulations, which are important for safety, security, and safeguards.



DNP groups approximate DNPs

A small number of artificial DNPs, known as DNP groups, are used to approximate the DNPs to reduce computational cost.

The three parameters for DNP groups are:

$\bar{\nu}_{d,k}$ = the delayed neutron yield of DNP group k [-]

τ_k = the half-life of DNP group k [s]

$\chi_{d,k}$ = the delayed neutron energy spectrum of DNP group k [-]

DNP groups are commonly used to approximate the DNPs in a light water reactor [3].



Safety, security, and safeguards are important

Safety, security, and safeguards by design is the act of developing a facility with features and processes that inherently address safety, security, and safeguards requirements from the earliest stages of design.

- *Safety* - Protect the public and minimize accident risk [14].
- *Security* - Prevent actors from acquiring information, stealing nuclear material, or committing sabotage [31].
- *Safeguards* - Impede the diversion or undeclared production of nuclear material for weapons by a host state [15].

Safety, security, and safeguards by design requires accurate transient simulations.

MSR phenomena are not captured in conventional parameters

Fluid-fueled molten salt reactor (MSR) phenomena that affect DNP group parameters:

- Recirculation of fuel salt
- Elemental chemical removal from reprocessing

The existing DNP group parameters do not account for MSR recirculation or chemical removal.

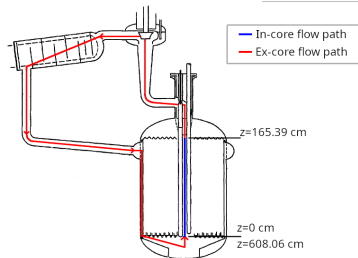


Figure 1: The flow path and bounds for the in-core region of the MSRE, overlayed on the MSRE loop diagram from [10].



This work will capture MSR phenomena in the DNP groups

The objectives of this work are to:

-  Create a Python package for DNP group parameter generation
-  Verify and validate the improved DNP group parameters
-  Evaluate the impact of these improved parameters on transient simulations for safety, security, and safeguards problems

The primary objective of this work is to generate improved DNP group parameters for DNPs.



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There are two approaches to calculating DNP group parameters

DNP group parameter calculation follows the following steps:

- ① Irradiate a fissile sample
- ② Measure/calculate a delayed neutron count rate
- ③ Determine the DNP group parameters that best fit that count rate

The specific steps depend on which approach is used.

Macroscopic approach

The delayed neutron count rate is *measured*.

Microscopic approach

The delayed neutron count rate is *calculated*.

The macroscopic approach irradiates a sample physically

The delayed neutron count rate ($\dot{n}_d$) is directly measured after sample irradiation.

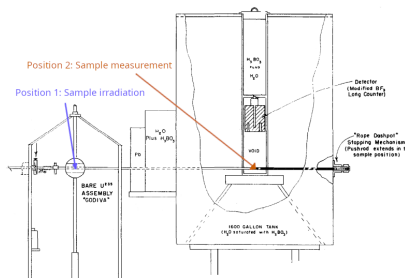


Figure 2: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [18].



The microscopic approach irradiates a sample computationally

The delayed neutron count rate is calculated from aggregated experimental data.

$$\dot{n}_d(t) = \sum_i^I P_{n,i} \lambda_i N_i(t), \quad (1)$$

where

I = the number of DNPs [#]

$\dot{n}_d$ = the delayed neutron count rate [$\# \cdot s^{-1}$]

$P_{n,i}$ = the average number of delayed neutrons emitted from decay of nuclide i , or the emission probability [#]

λ_i = the decay constant of DNP i [s^{-1}]

$N_i(t)$ = the number of atoms of DNP i [#].



There are two common irradiation types used to generate groups

The I DNPs are condensed into K groups.

Intermediate Irradiation

$$\dot{n}_d(T, t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \left(1 - e^{-\lambda_k T}\right) \quad (2)$$

Pulse irradiation ($T \rightarrow 0$)

$$\dot{n}_d(t) \approx F_s \sum_{k=1}^K \bar{\nu}_{d,k} \lambda_k e^{-\lambda_k t} \quad (3)$$

Saturation irradiation ($T \rightarrow \infty$)

$$\dot{n}_d(t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \quad (4)$$

The DNP parameters $\bar{\nu}_{d,k}$ and λ_k make $\dot{n}_d(t)$ match using exponential peeling [5, 17], least squares [4, 18], or half-lives of specific DNPs [27].

Fuel salt recirculation affects concentration of DNPs

Although normalized per fission, changes in the fission rate due to recirculation changes transmutation rates which could affect DNP concentrations.

In-core and ex-core salt recirculation can be modeled using:

- **Scaled flux method** [2]
- **Piecewise 0D method**
- Spatially resolved method [30, 28]

For a recirculating fissile sample, the piecewise 0D model offers a good balance of accuracy and speed.

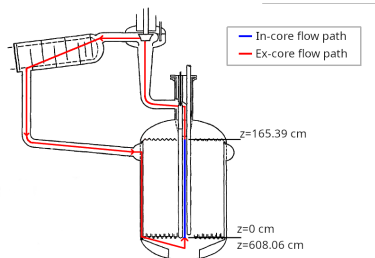


Figure 3: The flow path and bounds for the in-core region of the MSRE, overlaid on the MSRE loop diagram from [10].



Chemical reprocessing affects DNPs concentrations

Online chemical reprocessing includes:

- Fission product removal
- Fresh fuel addition

Conventional DNP group parameters do not account for these effects.

For example, a reprocessing scheme with 100% of all DNPs chemically removed should not have the same DNP groups.



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MoSDeN processes data and generates improved groups

Molten Salt Delayed Neutron (MoSDeN) is an open-source Python package in development that will enable users to generate improved DNP groups.

MoSDeN currently enables:

- Use of various data inputs for half-lives (τ), emission probabilities (P_n), and cumulative fission yields (CFY)
 - ENDF-6 formatted data
 - OpenMC chain data
 - Custom data
- Calculation of concentrations and delayed neutron count rates of an irradiated sample
- Calculation of the DNP group parameters that fit the calculated count rate

The MoSDeN package is in development and offers improved DNP group generation for MSRs on a per-problem basis.



The saturation irradiation is a non-linear least squares problem

The group parameters are calculated by equating the delayed neutron count rates.

$$\dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \approx \sum_i^I P_{n,i} \lambda_i N_i(t) \quad (5)$$

$$\dot{F}_s \begin{bmatrix} e^{-\lambda_1 t_0} & \dots & e^{-\lambda_K t_0} \\ \vdots & \ddots & \vdots \\ e^{-\lambda_1 t_m} & \dots & e^{-\lambda_K t_m} \end{bmatrix} \begin{bmatrix} \bar{\nu}_{d,1} \\ \vdots \\ \bar{\nu}_{d,K} \end{bmatrix} \approx \begin{bmatrix} \dot{n}_d(t_0) \\ \vdots \\ \dot{n}_d(t_m) \end{bmatrix} \quad (6)$$

$$A \vec{x} \approx \vec{b} \quad (7)$$



The residual best fits the delayed neutron count rate

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b}} \right\|_2^2, \quad (8)$$

where

$$\begin{aligned} \tau_k &\in [1 \text{ ms}, 1000 \text{ s}] \\ \vec{v}_{d,k} &\in [0, 1]. \end{aligned}$$

MoSDeN uses the SciPy package's optimize module to solve this non-linear least squares problem [33].



Uncertainties are tracked using Monte Carlo sampling

The process for uncertainty tracking is as follows:

- 1 Normally sample $P_{n,i}$, τ_i , and $N_i(t)$ within their uncertainties for every DNP
- 2 Calculate the DNP group parameters using non-linear least squares
- 3 Record the results and repeat

After all simulations finish:

- 1 Groups are defined based on half-lives (e.g. longest lived half-life and yield are group one)
- 2 The mean values and standard deviations are calculated



MoSDeN uses a newly-derived form of the group fit for MSRs

Assuming the sample experiences a square-wave power history during in-core and ex-core times, an MSR specific group fit equation can be derived.

$$\frac{F(\tau)}{F_0} = \sum_{j=0}^J H(\tau - j \cdot \tau_{net}) - H(\tau - j \cdot \tau_{net} - \tau_{in}) \quad (9)$$

where

$F(\tau)$ = the fission rate history $[\frac{\#}{s}]$.

H = the Heaviside function $[-]$

$$\tau_{net} = \tau_{in} + \tau_{ex}$$

= the total time per circulation $[s]$

$$J = \left\lfloor \frac{T}{\tau_{net}} \right\rfloor$$

= the number of recirculations $[\#]$

F_0 = the magnitude of the constant fission rate $[\frac{\#}{s}]$.



MoSDeN uses a newly-derived form of the group fit for MSRs

Assuming the sample experiences a square-wave power history during in-core and ex-core times, an MSR specific group fit equation can be derived.

$$\begin{aligned}
 \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{\nu}_{d,k}} = & \overbrace{e^{-\lambda_k t} \sum_{j=0}^{J_{in}} \left(e^{-\lambda_k (T-j \cdot \tau_{net} - \tau_{in})} - e^{-\lambda_k (T-j \cdot \tau_{net})} \right)}^{\text{Partial ex-core}} \\
 & + \underbrace{e^{-\lambda_k t} \sum_{j=J_{in}+1}^J \left(1 - e^{-\lambda_k (T-j \cdot \tau_{net})} \right)}_{\text{Partial in-core}}
 \end{aligned} \tag{10}$$

where

T = the total irradiation time

$$J_{in} = \left\lfloor \frac{T - \tau_{in}}{\tau_{net}} \right\rfloor$$

= the number of recirculated in-core irradiations [#]



The 0D scaled model is implemented in MoSDeN

MoSDeN uses simulation parameters and external data to generate improved DNP parameters.

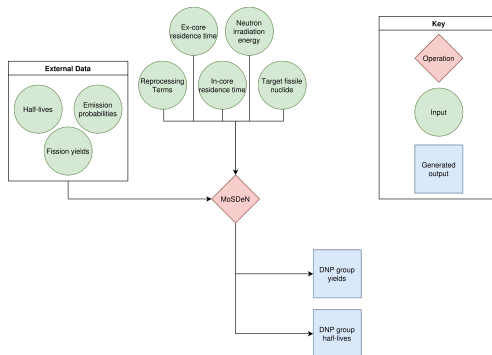


Figure 4: Flow diagram of 0D scaled model used to generate improved DNP group parameters.



The 0D scaled model is fast but uses many approximations

The equilibrium concentration of each DNP is calculated for the 0D scaled model as [13]:

$$N_i = \frac{f_{in} CFY_i}{\lambda_i}, \quad (11)$$

where

f_{in} = the fraction of time fuel-salt spends in the core [-]

CFY_i = the cumulative fission yield of nuclide i [-].

The limitations of the 0D model include:

- Concentration calculation does not include cross sections
- The concentration is only available at equilibrium
- The depletion chain is not tracked, so chemical removal is underestimated



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The parameters for the 0D scaled model analyses

Table 1: Default parameters for 0D scaled model simulations.

Parameter	Value	Units
Fissile Nuclide	^{235}U	[-]
Incident Neutron Energy	0.0253	[eV]
Fission Yields	ENDF/B-VII.1 [6]	[-]
Half-lives	IAEA [9]	[-]
Emission Probability	IAEA [9]	[-]
Final Decay Time	1200	[s]
Decay Time Spacing	Logarithmic	[-]
Decay Time Nodes	300	[-]
Sampled Simulations	1000	[-]
τ_{in}	9 [11]	[s]
τ_{ex}	16 [11]	[s]

The parameters for the 0D scaled model chemical removal analyses

Table 2: MSBR Online Reprocessing Removal Rates from [23].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

The total delayed neutron yield is dominated by a few DNPs

The delayed neutron yield from each DNP is given as [13, 19]:

$$\bar{\nu}_{d,i} = P_{n,i} C F Y_i. \quad (12)$$

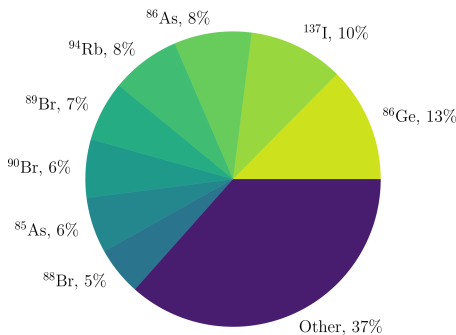


Figure 5: Relative delayed neutron yield from each DNP after a thermal saturation irradiation of ²³⁵U using the 0D scaled model.

The delayed neutron count rate is dominated by a few DNPs

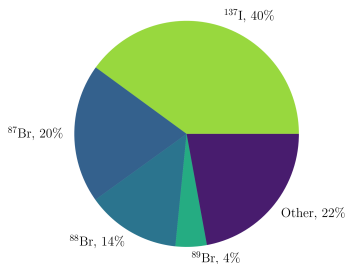


Figure 6: Relative delayed neutron count rates from each DNP after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

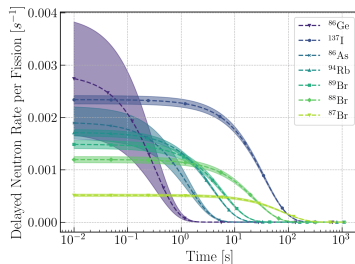


Figure 7: Relative delayed neutron counts after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Long cycle time elements have a negligible impact

Group half-lives:

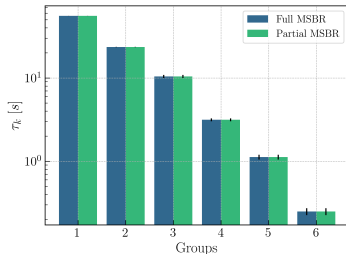


Figure 8: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

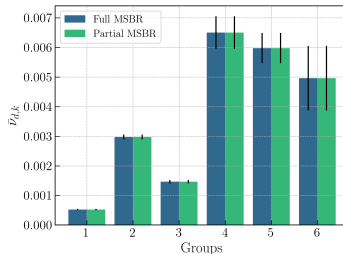


Figure 9: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

The differences induced by exclusion of removal rates with cycle times longer than 20 seconds are minimal and well within uncertainty bounds.

Removal rates have a small impact in the 0D scaled model

Group half-lives:

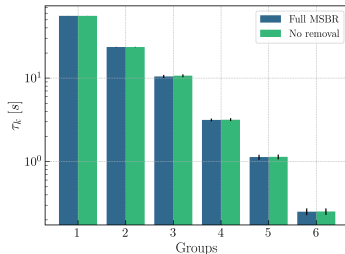


Figure 10: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

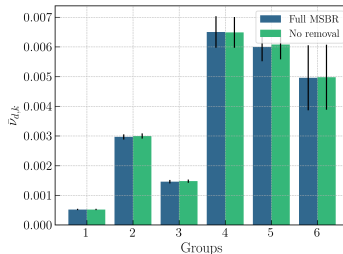


Figure 11: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Neglecting chemical removal rates generally leads to larger yields, though all differences are within uncertainty bounds.

Logarithmic spacing of time nodes give more accurate results

Group half-lives:

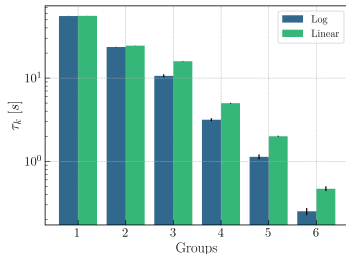


Figure 12: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

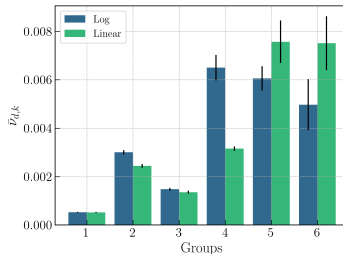


Figure 13: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Linear spacing has no data between zero and two seconds, and it places greater weight on longer-lived groups due to the data distribution.

More time nodes give slightly more accurate results

Group half-lives:

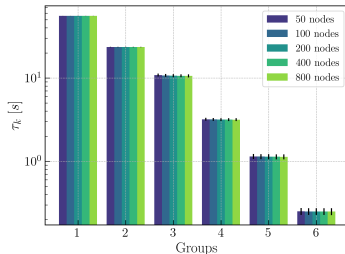


Figure 14: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

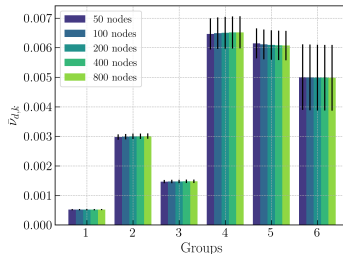


Figure 15: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Increasing the number of time nodes increases the accuracy of the DNP group parameters, with differences within uncertainty bounds.

Longer measurement times give more accurate results

Group half-lives:

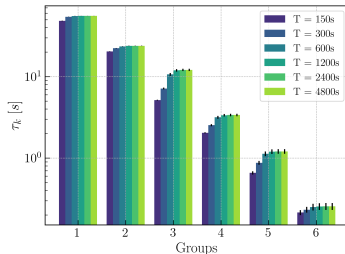


Figure 16: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

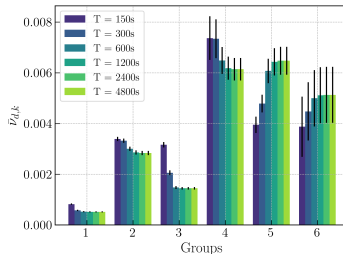


Figure 17: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Longer post-irradiation time results in longer-lived groups with smaller yields in groups one through four and larger yields in groups five and six.

The density of time nodes does not drive the group parameters

Group half-lives:

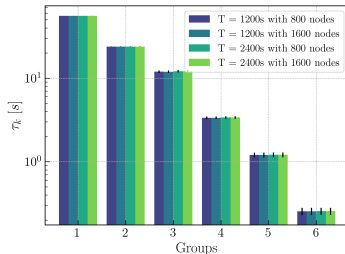


Figure 18: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

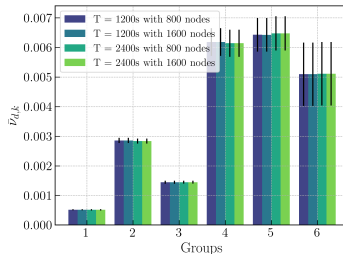


Figure 19: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

The number of nodes has a small impact on the DNP group parameters, whereas the total decay time causes larger differences.

PCCs convey sensitivity of DNP group parameters to nuclear data

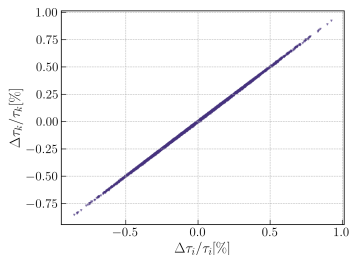


Figure 20: DNP group 1 half-life variation due to half-life variation with a PCC of 1.00.

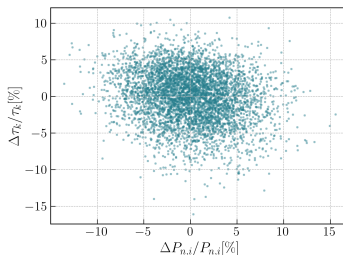


Figure 21: DNP group 3 half-life variation due to emission probability variation with a PCC of -0.19.

The half-life of ^{87}Br strongly correlates to the group one half-life with a Pearson Correlation Coefficient (PCC) magnitude close to 1.0, while its emission probability does not strongly correlate to the group three half-life.

The PCC magnitudes show the most sensitive DNPs

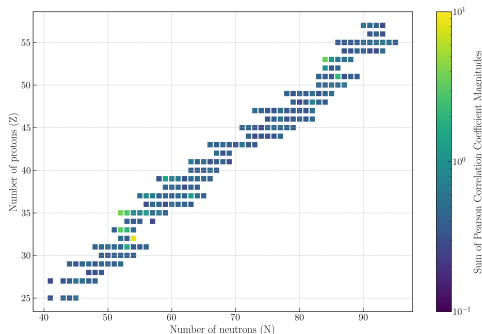


Figure 22: The summed PCC magnitudes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Table 3: The largest summed PCC magnitudes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nuclide	PCC_i
^{86}Ge	8.06
^{87}Br	3.82
^{137}I	3.52
^{85}As	3.14
^{88}Br	2.77
^{137}Sb	2.08
^{86}As	1.97
^{84}Ga	1.62
^{98}Y	1.42
^{100}Rb	1.22

The uncertainty-weighted PCC shows critical data needs

Uncertainty-weighted PCC U_i :

$$U_i = \sum_v \frac{\sigma_{i,v}}{x_{i,v}} \sum_g |PCC_{i,v,g}|, \quad (13)$$

where

$U_{i,v}$ = the uncertainty-weighted sensitivity of
DNP value v for DNP i [-]

$\sigma_{i,v}$ = the uncertainty in the DNP value v
for DNP i [-]

$x_{i,v}$ = the DNP value v , representing the half-life,
emission probability, or concentration of DNP i [-]

$PCC_{i,v,g}$ = the PCC for DNP i and value v with
DNP group parameter g [-].

Table 4: The largest summed uncertainty-weighted PCC magnitudes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nuclide	U_i
^{86}Ge	1.71
^{124}Ag	0.41
^{120}Ag	0.36
^{114}Tc	0.36
^{147}Xe	0.34
^{104}Nb	0.29
^{111}Mo	0.27
^{122}Pd	0.27
^{98}Y	0.26
^{103}Zr	0.26

The DNP group parameters accurately represent the DNP data

The total delayed neutron yield ($\bar{\nu}_d$) from the DNP group parameters should align with the DNP data:

$$\sum_{k=1}^K \bar{\nu}_{d,k} \approx \sum_{i=1}^I P_{n,i} CFY_i \quad (14)$$

The K groups give $\bar{\nu}_d = 2.24 \pm 0.13$ per 100 fissions, while the I DNPs give $\bar{\nu}_d = 2.25 \pm 0.15$.

The average half-life ($\bar{\tau}$) from the DNP group parameters should align with the DNP data:

$$\bar{\tau} = \ln(2) \sum_k^K \frac{\bar{\nu}_{d,k}}{\lambda_k}. \quad (15)$$

The K groups give $\bar{\tau} = 6.4 \pm 0.3s$, while the I DNPs give $\bar{\tau} = 6.4 \pm 0.4s$.

The total delayed neutron yield and average half-life agree between the model and the data.

The DNP group parameters do not align with the literature

The total delayed neutron yield ($\bar{\nu}_d$) from the DNP group parameters should align with the literature [21, 26, 6, 24, 20, 3, 7, 18, 8, 25, 29, 32, 34]:

The K groups give $\bar{\nu}_d = 2.24 \pm 0.13$ per 100 fissions, while the literature gives $1.48 \leq \bar{\nu}_d \leq 1.91$.

The average half-life ($\bar{\tau}$) from the DNP group parameters should align with the literature [24, 20, 3, 18, 8, 25, 29, 32, 34]:

The K groups give $\bar{\tau} = 6.4 \pm 0.3s$, while the literature gives $7.7 \leq \bar{\tau} \leq 9.5s$.

This indicates an issue with short-lived DNPs, likely from the selected nuclear data and the 0D model assumptions.

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The objectives of this work are to:

-  Create a Python package for DNP group parameter generation
-  Verify and validate the improved DNP group parameters
-  Evaluate the impact of these improved parameters on transient simulations for safety, security, and safeguards problems

The primary objective of this work is to generate improved DNP group parameters for DNPs.

The proposed work consists of three categories

Development of the 0D flow model:

- Enables improved DNP group parameters to account for residence time
- Includes development of the MoSDeN Python package
- Includes various other analyses and developments

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Development of the 0D flow model:

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Validation of the improved DNP group parameters:

- Ensures the accuracy of the 0D flow model
- Enables comparison between the 0D flow and 0D scaled models
- Validates the accuracy of the MSRE transient simulation using Moltres

The proposed work consists of three categories

Development of the 0D flow model:

- Enables improved DNP group parameters to account for residence time
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Validation of the improved DNP group parameters:

- Ensures the accuracy of the 0D flow model
- Enables comparison between the 0D flow and 0D scaled models
- Validates the accuracy of the MSRE transient simulation using Moltres

Analysis of safety, security, and safeguards using MSRE transients:

- Demonstrates the effects of the DNP group parameters
- Conventional DNP parameters may predict differing behavior than the improved parameters



The 0D flow model will introduce new features and analyses

New features in the 0D flow model:

- Full depletion chain modeling in OpenMC
 - Improving chemical removal modeling
 - Enabling residence time analysis
 - Enabling non-equilibrium DNP concentrations
 - Improving DNP concentration accuracy
- Novel analysis for residual minimization using pulse and saturation irradiation data [4, 18]
- Comparison of various nuclear datasets [13, 21, 19]
- Calculation of the DNP group spectra ($\chi_{d,k}$)



The MSRE will be used for validation

The validation will proceed as follows:

- Calculation of improved DNP group parameters for MSRE
- K-eigenvalue calculations of MSRE for multi-group cross sections
- Add approximate models of upper and lower plena to existing MSRE Moltres model
- Calculation of β_{eff} using the prompt method for stationary and flowing fuel salt using improved and conventional DNP group parameters
- Comparison of β_{eff} with experimental data [12]
- Simulate pump start-up and coast-down transients
- Comparison of reactivity response with experimental data [22]



Safety-related transients may disrupt safe reactor operation

Safety-related transients that will be modeled are:

- Unprotected loss of heat sink (ULOHS)
 - Loss of cooling in the reactor due, potentially due to a heat exchanger failure
 - Modeled in Moltres by setting inlet temperature equal to outlet temperature
- Unprotected loss of flow (ULOF)
 - Flow rate decreases to zero due to a pump failure
 - Modeled in Moltres by changing flow field over time to match MSRE pump shut-down
- Unprotected transient overpower (UTOP)
 - Reactivity insertion event, which could be caused by poison removal, fissile addition, or temperature change
 - Modeled in Moltres by having a step-decrease in the removal cross section, representing control rod withdrawal
- Fuel salt overcooling (FSOC)
 - Fuel salt temperature decreases, potentially caused by increased heat removal rate
 - Modeled in Moltres by reducing inlet temperature by 25%

Security-related transients from sabotage may be used as a diversion

This framework assumes a hostile actor and an insider with the objective to disrupt operations [31].

Sabotage-initiated transients that will be modeled are:

- All safety-related transients as single events
- Combined UTOP and ULOHS
 - This should cause a temperature increase that propagates due to lack of heat sink; reactivity should then decrease due to temperature increase
- Combined UTOP and ULOF
 - This should cause a temperature increase with a peak maximum temperature due to stationary fuel salt; reactivity should then decrease due to temperature increase
- Combined UTOP and FSOC
 - This should cause a temperature increase that is counteracted by the overcooled inlet fuel salt, leading to the largest reactivity change; temperature is not expected to change dramatically

SNM diversion may be detected using reactor kinetics

The DNP group parameters will then be used in a negative reactivity step insertion, and the resulting power responses will be compared [35].

Special nuclear material (SNM) diversion scenarios considered are [31, 16, 1]:

- One significant quantity (SQ) of ^{235}U (75 kg) after one year in the MSRE
 - Removing ^{235}U will decrease the delayed neutron fraction
- One SQ of Pu (8 kg) after one year in the MSRE
 - Removing Pu will increase the delayed neutron fraction

Each scenario will have four sets of DNP group parameters to compare:

- Conventional parameters with/without diversion
- Improved parameters with/without diversion

The change in fuel composition and the MSR phenomena affect the ability to detect SNM diversion.

Summary

The primary objective of this work is to generate improved DNP group parameters for DNPs.

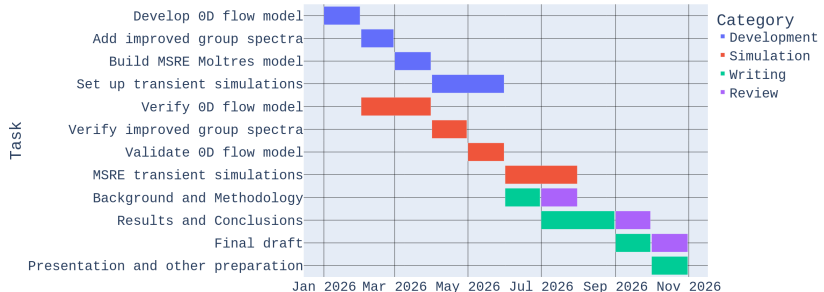


Figure 23: A Gantt chart of the proposed work.



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